BBM126- Programming forData Analytics

ExploratoryData Analysis (EDA) Project

Candidate No: 250301807

Introduction

This project conducts an exploratory data analysis (EDA) to better understand the factors associated with variations in hourly wages within a sample of individuals. The aim is to examine how demographic characteristics (such as gender, race and marital status) and human-capital variables (such as education and employment experience) relate to wage differences. These variables have practical implications for understanding labour-market inequalities, income disparities and the role of education and experience in shaping earning potential. Following a structured sequence of analytical steps, the report begins by preparing and summarising the dataset, handling missing values and visualising key distributions. Statistical tests are then used to assess relationships between variables, while subsetting and grouping techniques provide deeper insight into specific demographic patterns. Finally, a linear regression model is developed to evaluate how multiple variables jointly influence wages. Together, these steps aim to build a clear and evidence-based understanding of wage determinants in the dataset.

In [172…

```
#Importing the relevant libraries

#General packages
import pandas as pd
import numpy as np

#For visualisations
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

#For statistical analysis
from scipy.stats import chi2_contingency, ttest_ind
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Supressed warnings for easier readability
import warnings
warnings.filterwarnings('ignore')
```

In [173…
```
# Load the data
df = pd.read_excel("wage.xlsx")
```

Descriptive Statistics

Generate descriptive statistics for the dataset, and comment on the main trends.

In [174…
```
# View the data (first five rows)
print("First 5 rows:")
display(df.head())
```

```
First 5 rows:
```

	married	hourly_wage	years_in_education	years_in_employment	num_dependents	gender	race
0	1.0	3.24	12.0	2.0	3.0	female	white
1	0.0	3.00	11.0	0.0	2.0	male	white
2	1.0	6.00	8.0	28.0	0.0	male	white
3	1.0	5.30	12.0	2.0	1.0	male	white
4	1.0	8.75	16.0	8.0	0.0	male	white

In [175…
```
#View data types and counts
print("\nInfo:")
df.info()
```

```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525 entries, 0 to 524
Data columns (total 7 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   married             522 non-null    float64
 1   hourly_wage         517 non-null    float64
 2   years_in_education  522 non-null    float64
 3   years_in_employment 519 non-null    float64
 4   num_dependents      520 non-null    float64
 5   gender              521 non-null    object 
 6   race                515 non-null    object 
dtypes: float64(5), object(2)
memory usage: 28.8+ KB
```

Based on the above output, the data types of each column are summarised below: Numerical Continuous: married, hourly_wage, years_in_education, years_in_employment Categorical Nominal: gender, race

In [176…

```python
#view descriptive summary for each variable
print("\nSummary statistics:")
df.describe(include='all')
```

```
Summary statistics:
```

Out[176…

	married	hourly_wage	years_in_education	years_in_employment	num_dependents	gender	race
count	522.000000	517.000000	522.000000	519.000000	520.000000	521	515
unique	NaN	NaN	NaN	NaN	NaN	2	2
top	NaN	NaN	NaN	NaN	NaN	male	white
freq	NaN	NaN	NaN	NaN	NaN	272	461
mean	0.609195	5.917737	12.557471	5.152216	1.044231	NaN	NaN
std	0.488399	3.699058	2.757219	7.257133	1.258484	NaN	NaN
min	0.000000	0.530000	0.000000	0.000000	0.000000	NaN	NaN
25%	0.000000	3.350000	12.000000	0.000000	0.000000	NaN	NaN
50%	1.000000	4.670000	12.000000	2.000000	1.000000	NaN	NaN
75%	1.000000	6.880000	14.000000	7.000000	2.000000	NaN	NaN
max	1.000000	24.980000	18.000000	44.000000	6.000000	NaN	NaN

Intrepretation: The dataset consists of 525 records of different individuals across 7 columns: married, hourly_wage, years_in_education, years_in_employment, num_dependents, gender and race. The main trends are: - The mean hourly_wage is 5.92 with a standard deviation of +/-3.7 which is quite high. The minimum hourly wage is 0.53 while the maximum is 24.98. The median is 4.67 which suggests that this variable is slightly skewed to the right due to high outlier hourly wages. - The mean years_in_education for the individuals is 12.55 with a standard deviation of +/-2.75. The minimum years in education is 0 years while the maximum is 18 years. The median is 12 which is close to the mean. - The mean years_in_employment for the individuals is 5.15 with a standard deviation of +/-7.25 which is again quite high. The minimum years in employment is 0 years while the maximum is 44 years. The median is 2 which means data is highly skewed towards the right. - the mean no_of_dependants is 1.04 with standard deviation of 1.25. The mimimum dependants are 0 and maximum dependants are 6. The median is 1 which is close to the mean. - The married variable although categorised as a continuous variable shows only 0 and 1 values for unmarried and married respectively. The mean being 0.6 shows there are more married individuals in the dataset. - The gender variable shows only 2 unique values representing 2 genders. The mode being male with a count of 272 shows there are more males than females in the dataset. - The race variable shows only 2 unique values representing white and non-white. The mode being white with a count of 461 shows there are more whites than non-whites in the dataset.

Missing Values Analysis

Check any records with missing values and handle the missing data as appropriate.

In [177…

```
#Count missing values
print("\nMissing Value Counts:")
df.isnull().sum()
```

```
Missing Value Counts:
```

Out[177…

```
married                3
hourly_wage            8
years_in_education     3
years_in_employment    6
num_dependents         5
gender                 4
race                  10
dtype: int64
```

In [178…

```
#Count unique values
print("\nUnique Value Counts:")
for col in df.columns:
    print(f"\n{col}")
    print("\nValue counts:")
    print(df[col].value_counts(dropna=False))
```

```
Unique Value Counts:

married

Value counts:
married
1.0    318
0.0    204
NaN      3
Name: count, dtype: int64

hourly_wage

Value counts:
hourly_wage
3.00     33
4.50     20
6.25     18
3.50     17
2.90     16
         ..
5.52      1
2.31      1
2.83      1
8.65      1
11.56     1
Name: count, Length: 238, dtype: int64

years_in_education

Value counts:
years_in_education
12.0    198
16.0     67
14.0     53
13.0     39
10.0     30
11.0     28
8.0      21
15.0     21
18.0     18
9.0      17
17.0     12
6.0       6
7.0       4
```

```
4.0       3
NaN       3
0.0       2
2.0       1
3.0       1
5.0       1
Name: count, dtype: int64

years_in_employment

Value counts:
years_in_employment
0.0     159
2.0      62
1.0      51
3.0      41
5.0      30
4.0      27
8.0      15
7.0      15
6.0      14
10.0     13
9.0       9
11.0      9
12.0      8
13.0      8
NaN       6
15.0      6
21.0      5
20.0      5
24.0      5
16.0      4
25.0      4
30.0      3
23.0      3
18.0      3
17.0      3
26.0      3
31.0      2
22.0      2
28.0      2
19.0      2
14.0      2
44.0      1
39.0      1
```

```
34.0      1
33.0      1
Name: count, dtype: int64

num_dependents

Value counts:
num_dependents
0.0    248
1.0    105
2.0     98
3.0     45
4.0     15
5.0      7
NaN      5
6.0      2
Name: count, dtype: int64

gender

Value counts:
gender
male      272
female    249
NaN         4
Name: count, dtype: int64

race

Value counts:
race
white       461
nonwhite     54
NaN          10
Name: count, dtype: int64
```

In [179…

```
# impute missing values
# Binary/category variables → MODE
cat_vars = ['married', 'gender', 'race']

for col in cat_vars:
    df[col] = df[col].fillna(df[col].mode()[0])

# Skewed numeric variables → MEDIAN
num_vars = ['hourly_wage', 'years_in_education', 'years_in_employment', 'num_dependents']
```

```
for col in num_vars:
    df[col] = df[col].fillna(df[col].median())
```

In [180…

```
#view descriptive summary for each variable after imputation
print("\nSummary statistics (numeric columns):")
df.describe(include='all')

df.info()
```

```
Summary statistics (numeric columns):
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525 entries, 0 to 524
Data columns (total 7 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   married             525 non-null    float64
 1   hourly_wage         525 non-null    float64
 2   years_in_education  525 non-null    float64
 3   years_in_employment 525 non-null    float64
 4   num_dependents      525 non-null    float64
 5   gender              525 non-null    object 
 6   race                525 non-null    object 
dtypes: float64(5), object(2)
memory usage: 28.8+ KB
```

In [181…

```
# correct data types
df['married'] = df['married'].astype(int)
df['years_in_education'] = df['years_in_education'].astype(int)
df['years_in_employment'] = df['years_in_employment'].astype(int)
df['num_dependents'] = df['num_dependents'].astype(int)
df['gender'] = df['gender'].astype('category')
df['race'] = df['race'].astype('category')

#view again
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525 entries, 0 to 524
Data columns (total 7 columns):
 #   Column               Non-Null Count  Dtype   
---  ------               --------------  -----   
 0   married              525 non-null    int64   
 1   hourly_wage          525 non-null    float64 
 2   years_in_education   525 non-null    int64   
 3   years_in_employment  525 non-null    int64   
 4   num_dependents       525 non-null    int64   
 5   gender               525 non-null    category
 6   race                 525 non-null    category
dtypes: category(2), float64(1), int64(4)
memory usage: 21.9 KB
```

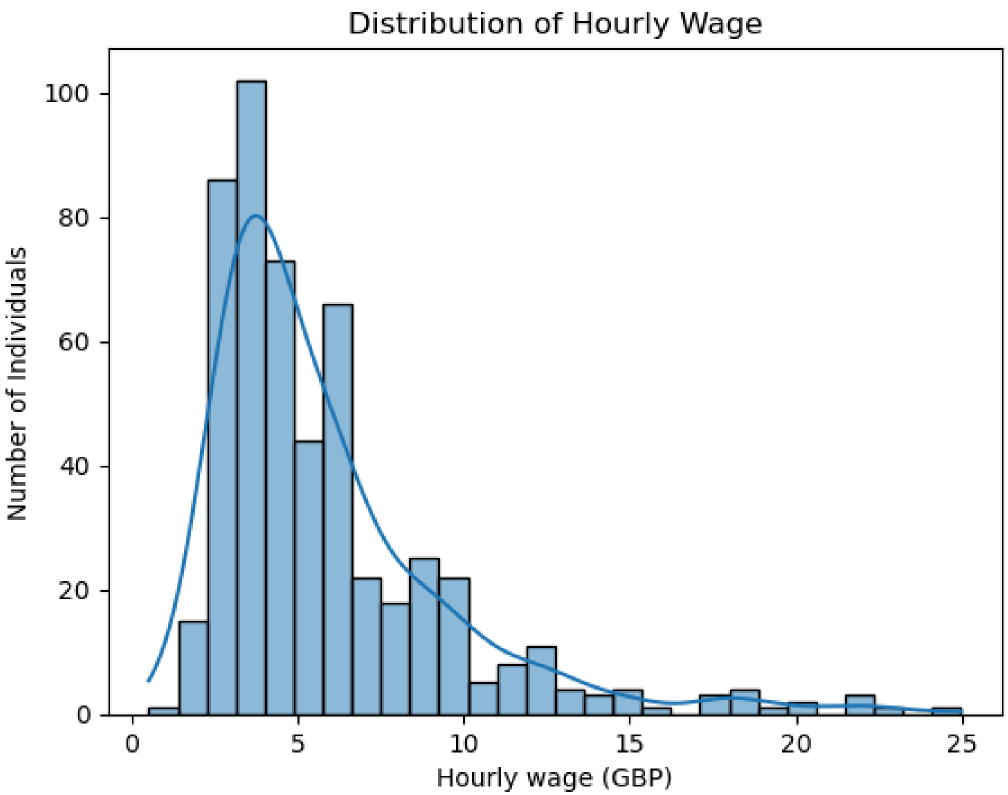
Intrepretation: A review of the dataset showed that several variables contained missing values, with counts ranging from 3 to 10 records per variable. Before conducting any statistical analysis, these missing values were handled using imputation techniques appropriate to the type and distribution of each variable. - Binary and Categorical Variables (Imputed Using Mode) The variables married, gender, and race are categorical in nature. married is a binary indicator (0 = not married, 1 = married). gender consists of the categories male and female. race contains two population groups: white and non-white. For categorical variables, the mode (the most frequent category) is the most suitable method of imputation. This approach preserves the underlying distribution of categories without introducing artificial numeric values. Imputing with the mean or median would be inappropriate, as these variables do not represent continuous quantities. Therefore, all missing values in married, gender, and race were replaced with their respective mode categories. - Numeric Variables (Imputed Using Median Due to Skewness) The variables hourly_wage, years_in_education, years_in_employment, and num_dependents are numeric, but all exhibit skewed distributions. Because the mean is sensitive to extreme values and skewness, using it could distort the underlying variable structure and produce unrealistic imputed values. In addition, these numerical values consist of integer values only. The median provides a more robust and representative central value for skewed data while ensuring (mostly) integer values for the missing data.

Data Visualisation

Build graphs visualizing the following and comment on the obtained visual insights A. the distribution of one or more individual continuous variables B. the relationship of a pair of continuous variables C. the association b/w a categorical variable and a continuous one D. The relationship between more than two variables

In [182…

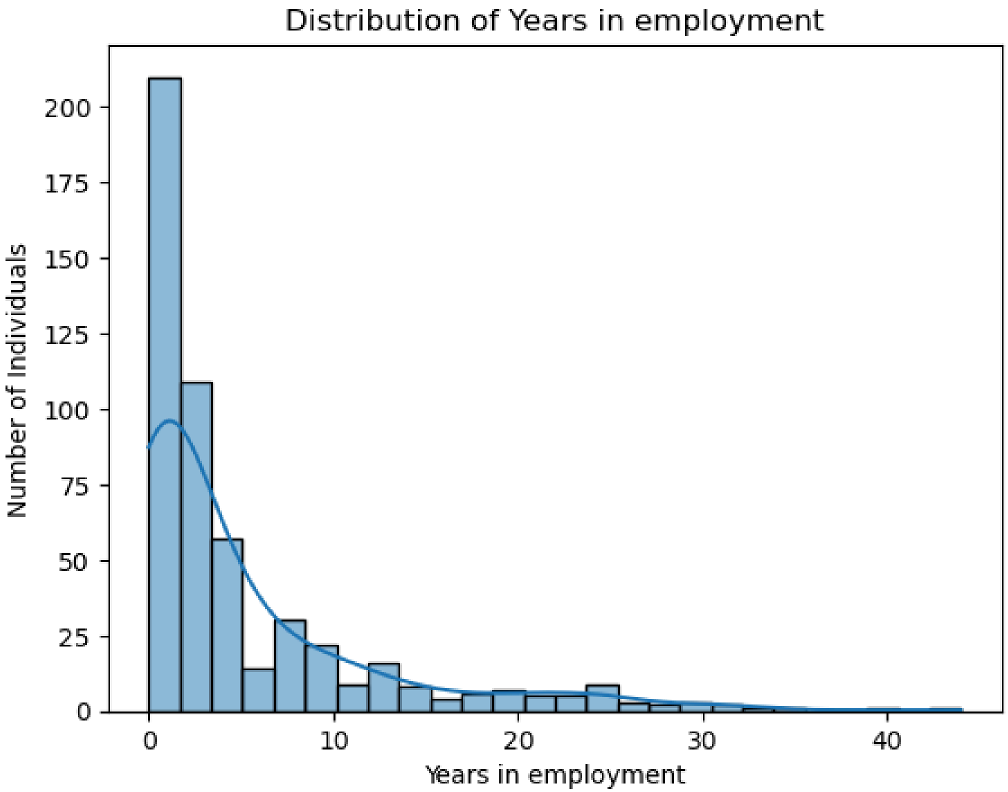
```
# A. Distributions of individual continuous variables
# 1. histogram of hourly wage
sns.histplot(df['hourly_wage'].dropna(), kde='true')
plt.title("Distribution of Hourly Wage")
plt.ylabel("Number of Individuals")
plt.xlabel("Hourly wage (GBP)")
plt.show()
```



The distribution of hourly wages is right-skewed, with most individuals earning between £3 and £7 per hour. A smaller number earn higher wages, and very few individuals earn above £15 per hour, creating a long tail on the right side of the distribution. This suggests that while lower and mid-range wages are common, high-wage earners are relatively rare in the dataset.

In [183…

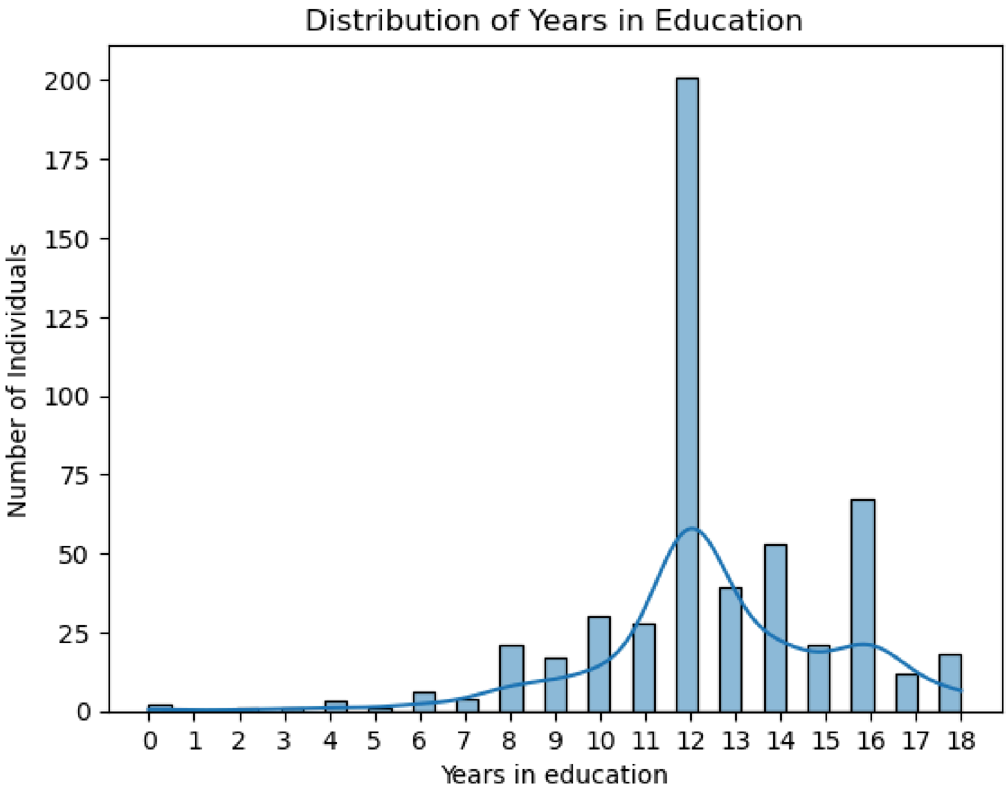
```
# A. Distributions of individual continuous variables
# 2. histogram of years in employment
sns.histplot(df['years_in_employment'].dropna(), kde='true')
plt.title("Distribution of Years in employment")
plt.xlabel("Years in employment")
plt.ylabel("Number of Individuals")
plt.show()
```



The distribution of years in employment is highly right-skewed, with most individuals having between 0 and 10 years of employment experience. A large concentration is seen at very low values, including many individuals with only a few years of experience. Only a small number of participants have long employment histories (20+ years), creating a long tail on the right. This suggests the dataset is dominated by individuals early in their careers, with relatively few long-tenured workers.

In [184…

```
# A. Distributions of individual continuous variables
# 3. histogram of years in education
sns.histplot(df['years_in_education'].dropna(), kde='true')
plt.title("Distribution of Years in Education")
plt.xlabel("Years in education")
plt.ylabel("Number of Individuals")
# Force x axis to use step of 1
xmin = int(df['years_in_education'].min())
xmax = int(df['years_in_education'].max())
plt.xticks(range(xmin, xmax + 1, 1))
plt.show()
```



The distribution of years in education is slightly left-skewed, with most individuals completing between 10 and 16 years of education. The tallest bar appears around 12 years, suggesting that secondary education completion is the most common level in the dataset. A smaller number of individuals have very low (0–6 years) or very high (17–18 years) levels of education, indicating that extreme values are relatively rare. Overall, the distribution is fairly concentrated around the middle range of schooling years.

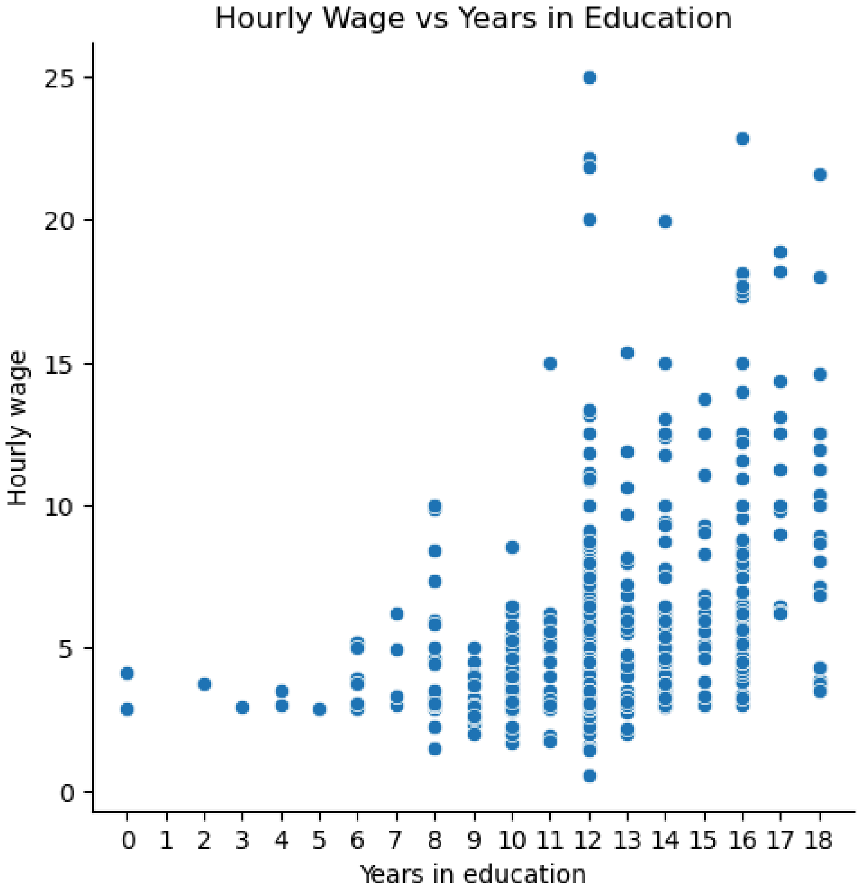
In [185…

```
# B. the relationship of a pair of continuous variables

# Hourly wage against years in education
plt.figure(figsize=(6,4))
sns.relplot(
    x="years_in_education",
    y="hourly_wage",
    data=df)
plt.title("Hourly Wage vs Years in Education")
plt.xlabel("Years in education")
plt.ylabel("Hourly wage")
# Force x axis to use step of 1
xmin = int(df['years_in_education'].min())
xmax = int(df['years_in_education'].max())
```

```
plt.xticks(range(xmin, xmax + 1, 1))
plt.show()
```

```
<Figure size 600x400 with 0 Axes>
```



The scatterplot shows a positive relationship between years in education and hourly wage. Individuals with fewer than 10 years of education tend to earn lower wages, mostly below £7 per hour. As years in education increase particularly beyond 12 years both the typical wage and the range of possible wages increase. Higher education levels (15–18 years) are associated with a wider spread and more frequent higher-wage observations, including wages above £15 and up to £25 per hour. Although the relationship is not perfectly linear and there is considerable variation at each education level, the overall pattern suggests that higher educational attainment is generally linked to higher earning potential.

In [186…

```
# Hourly wage against years in employment
plt.figure(figsize=(6,4))
sns.relplot(
    x="years_in_employment",
    y="hourly_wage",
    data=df
)
```

```
plt.title("Hourly Wage vs Years in Employment")
plt.xlabel("Years in employment")
plt.ylabel("Hourly wage")
plt.show()
```

```
<Figure size 600x400 with 0 Axes>
```

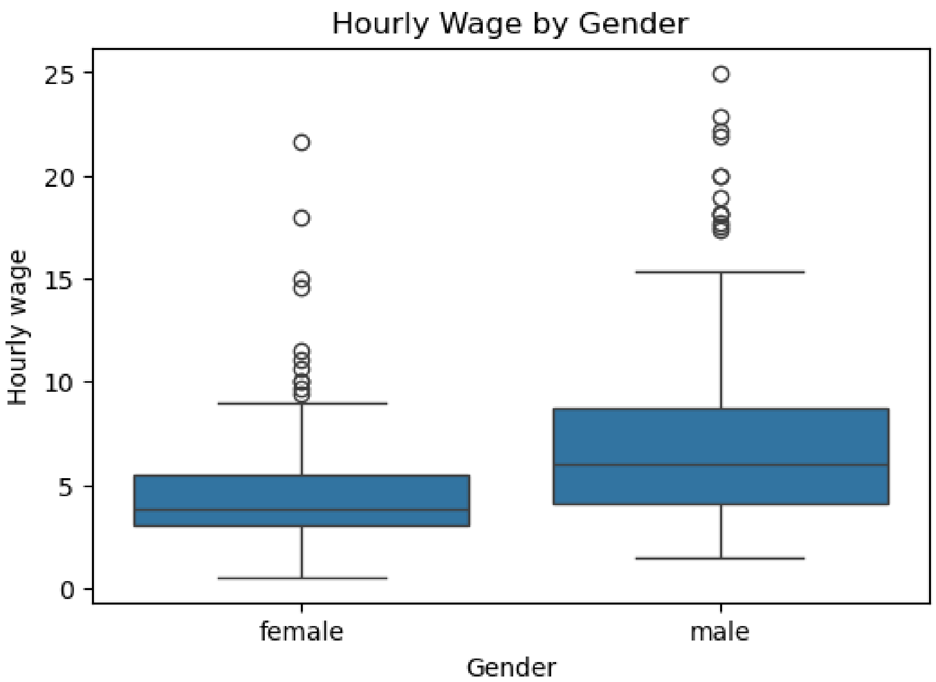
The scatterplot shows a positive but weak relationship between years in employment and hourly wage. Most individuals have fewer than 10 years of employment and tend to earn lower wages, typically between £3 and £8 per hour. As employment experience increases—particularly beyond 10–15 years—the range of wages also increases, and higher wages (above £15–£20 per hour) become more common. However, the data points are widely scattered, indicating substantial variability in wages at all levels of employment experience. Even among individuals with many years of experience (20–40 years), wages vary significantly, suggesting that employment experience influences wages but is not the sole determinant. Many other factors (such as education, occupation, hours worked) likely contribute to wage differences.

In [187…

```
# C. the association b/w a categorical variable and a continuous one

plt.figure(figsize=(6,4))
sns.boxplot(
    x="gender",
```

```
    y="hourly_wage",
    data=df
)
plt.title("Hourly Wage by Gender")
plt.xlabel("Gender")
plt.ylabel("Hourly wage")
plt.show()
```

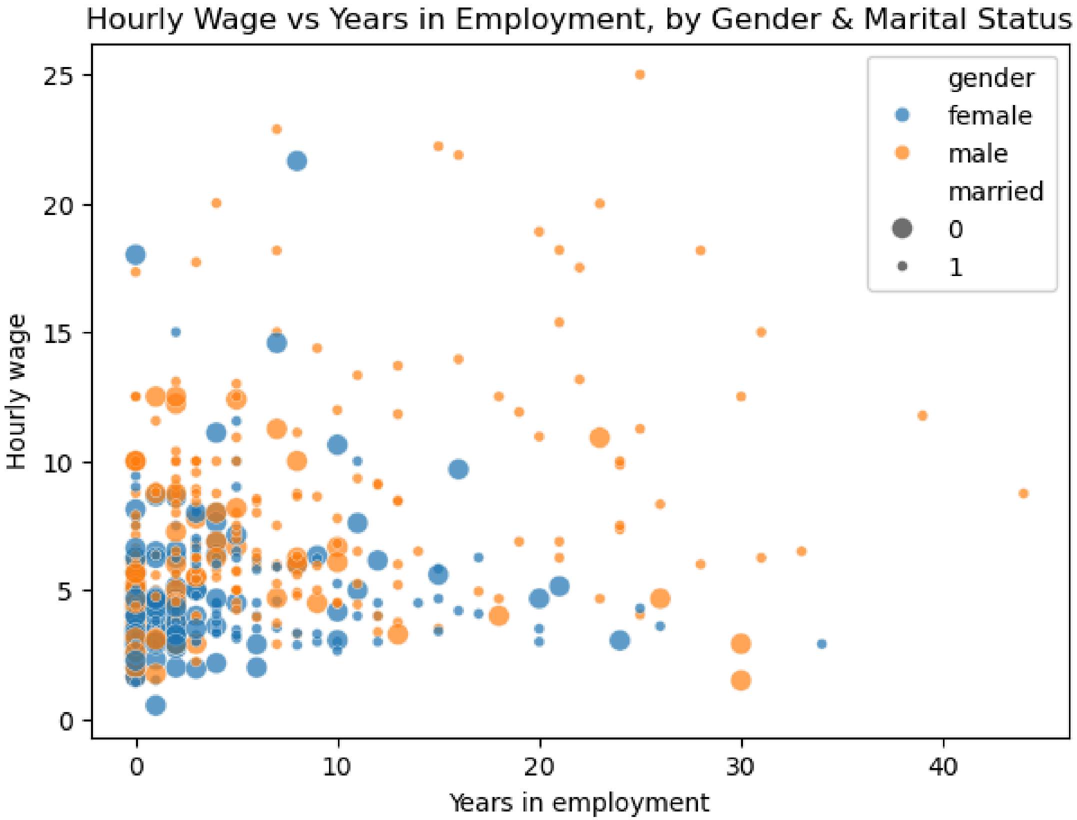


The boxplot shows a clear difference in hourly wages between males and females. Male workers have a noticeably higher median hourly wage compared with female workers, and their entire wage distribution is shifted upward. Males also display a wider spread and a larger number of high-wage outliers, including values above £15 and even above £20 per hour. In contrast, the female wage distribution is lower and more concentrated, with fewer high-wage observations and a lower upper quartile. Overall, the visual pattern indicates that men tend to earn higher hourly wages than women, and higher-paying jobs appear more common among men in this dataset.

In [188…

```
#D. The relationship between more than two variables
# scatter of wage vs education, coloured by gender, size by years_in_employment
plt.figure(figsize=(7,5))
sns.scatterplot(
    data=df,
    x="years_in_employment",
    y="hourly_wage",
    hue="gender",
    size="married",
    alpha=0.7
```

```
)
plt.title("Hourly Wage vs Years in Employment, by Gender & Marital Status")
plt.xlabel("Years in employment")
plt.ylabel("Hourly wage")
plt.legend()
plt.show()
```



This plot shows how hourly wages vary with employment experience, while also distinguishing individuals by gender (colour) and marital status (bubble size). 1. Gender differences Male workers (orange) consistently appear at higher wage levels, especially for wages above £15. Female workers (blue) are more concentrated in the lower wage ranges, even at similar levels of employment experience. This reinforces the observed gender wage gap. 2. Employment experience Wages tend to increase with years in employment, but the relationship is not strong. Even among individuals with many years of experience (20+ years), wages vary widely. 3. Marital status Larger bubbles represent married individuals. Many of the higher-wage points belong to married individuals, particularly married men. Unmarried individuals (smaller bubbles) appear more concentrated in lower-wage ranges. Overall pattern The chart suggests that gender and marital status both contribute to wage differences beyond what can be explained by years in employment alone. Men—especially married men—are more represented among higher-wage earners, while women and unmarried individuals cluster in lower to mid-range wages.

Categorical Variable Analysis

Display unique values of a categorical variable and their frequencies.

In [189…

```
# Categorical variable summary stats
for col in ['gender', 'married', 'race']:
    print(f"\n{col.upper()}")
    print("Unique values:")
    print(df[col].unique())

    print("\nFrequencies:")
    print(df[col].value_counts(dropna=False))
```

```
GENDER
Unique values:
['female', 'male']
Categories (2, object): ['female', 'male']

Frequencies:
gender
male      276
female    249
Name: count, dtype: int64

MARRIED
Unique values:
[1 0]

Frequencies:
married
1    321
0    204
Name: count, dtype: int64

RACE
Unique values:
['white', 'nonwhite']
Categories (2, object): ['nonwhite', 'white']

Frequencies:
race
white       471
nonwhite     54
Name: count, dtype: int64
```

The majority of the individuals in the database were married, white and male.

Contingency Table

Build a contingency table of two potentially related categorical variables, conduct a statistical test of the independence between them, then interpret the results.

In [190…

```
# Contingency table: gender vs married
contingency_table = pd.crosstab(df['gender'], df['married'])
print("Contingency Table (Gender vs Married):")
display(contingency_table)
```

```
Contingency Table (Gender vs Married):
```

married	0	1
gender		
female	118	131
male	86	190

In [191…

```
# contingency table results

chi2, p, dof, expected = chi2_contingency(contingency_table)

print("\nChi-square statistic:", round(chi2, 4))
print("Degrees of freedom:", dof)
print("p-value:", round(p, 4))

# Show expected frequencies
expected_df = pd.DataFrame(expected,
                           index=contingency_table.index,
                           columns=contingency_table.columns)

print("\nExpected Frequencies:")
display(expected_df)
```

```
Chi-square statistic: 13.8386
Degrees of freedom: 1
p-value: 0.0002

Expected Frequencies:
```

married	0	1
gender		
female	96.754286	152.245714
male	107.245714	168.754286

Interpretation: A chi-square test of independence was conducted to examine the relationship between gender and marital status. The test produced a chi-square statistic of 13.84 with 1 degree of freedom and a p-value of 0.000199. Since the p-value is well below the 0.05 significance threshold, the null hypothesis of independence is rejected. This indicates that gender and marital status are significantly associated in the dataset. In practical terms, the probability of being married differs between males and females. All expected cell frequencies were greater than five, confirming that the assumptions of the chi-square test were met.

In [192…

```
## The normalize argument is used to convert values into proportions of the whole
proportions = pd.crosstab(
    df['married'],
    df['gender'],
    normalize='index') * 100  # Convert to percentage

print('Marital Status by Gender')
print(proportions)
```

```
Marital Status by Gender
gender      female       male
married
0        57.843137  42.156863
1        40.809969  59.190031
```

This shows that males are mostly married and females are mostly unmarried in our dataset.

Data Subsetting

Retrieve one or more subsets of rows based on two or more logical criteria and present descriptive statistics on the subset(s).

We are going to create two subsets of males and females who are both highly educated and have high number of years of experience in order to compare them later on.

In [193…

```
# calculate medians to filter out high education and high experience (due to skewness identified earlier)
median_edu = df['years_in_education'].median()
median_emp = df['years_in_employment'].median()

# Subset 1: females who have high education and high experience
subset_female_highedu_highexp = df[
    (df['gender'] == 'female') &
    (df['years_in_education'] >= median_edu) &
    (df['years_in_employment'] >= median_emp)]

print("\nNumber of observations in the female + high edu + high exp:", len(subset_female_highedu_highexp))
```

```
display(subset_female_highedu_highexp.describe())

# Subset 2: males who have high education and high experience
subset_male_highedu_highexp = df[
    (df['gender'] == 'male') &
    (df['years_in_education'] >= median_edu) &
    (df['years_in_employment'] >= median_emp)]

print("\nNumber of observations in the male + high edu + high exp:", len(subset_male_highedu_highexp))
display(subset_male_highedu_highexp.describe())
```

```
Number of observations in the female + high edu + high exp: 117
```

	married	hourly_wage	years_in_education	years_in_employment	num_dependents
count	117.000000	117.000000	117.000000	117.000000	117.000000
mean	0.615385	5.342821	13.264957	6.196581	0.880342
std	0.488597	2.827579	1.657757	5.513681	1.043555
min	0.000000	2.000000	12.000000	2.000000	0.000000
25%	0.000000	3.450000	12.000000	2.000000	0.000000
50%	1.000000	4.500000	12.000000	4.000000	1.000000
75%	1.000000	6.250000	14.000000	8.000000	2.000000
max	1.000000	21.630000	18.000000	34.000000	4.000000

```
Number of observations in the male + high edu + high exp: 139
```

	married	hourly_wage	years_in_education	years_in_employment	num_dependents
count	139.000000	139.000000	139.000000	139.000000	139.000000
mean	0.805755	8.939784	14.028777	9.007194	1.086331
std	0.397049	4.502514	2.053426	8.449763	1.176349
min	0.000000	2.920000	12.000000	2.000000	0.000000
25%	1.000000	5.840000	12.000000	3.000000	0.000000
50%	1.000000	8.000000	14.000000	5.000000	1.000000
75%	1.000000	10.935000	16.000000	12.500000	2.000000
max	1.000000	24.980000	18.000000	44.000000	4.000000

In [194…

```
#total descriptive stats for comparison
print("\nSummary statistics:")
df.describe(include='all')
```

```
Summary statistics:
```

Out[194…

	married	hourly_wage	years_in_education	years_in_employment	num_dependents	gender	race
count	525.000000	525.000000	525.000000	525.000000	525.000000	525	525
unique	NaN	NaN	NaN	NaN	NaN	2	2
top	NaN	NaN	NaN	NaN	NaN	male	white
freq	NaN	NaN	NaN	NaN	NaN	276	471
mean	0.611429	5.898724	12.554286	5.116190	1.043810	NaN	NaN
std	0.487890	3.673900	2.749637	7.223254	1.252473	NaN	NaN
min	0.000000	0.530000	0.000000	0.000000	0.000000	NaN	NaN
25%	0.000000	3.350000	12.000000	0.000000	0.000000	NaN	NaN
50%	1.000000	4.670000	12.000000	2.000000	1.000000	NaN	NaN
75%	1.000000	6.880000	14.000000	7.000000	2.000000	NaN	NaN
max	1.000000	24.980000	18.000000	44.000000	6.000000	NaN	NaN

Interpretation: As expected, the average years of education for subset 1 and subset 2 are 13.26 and 14.02 which are both higher than the average in dataset 12.55 as we filtered based on this criteria. The average for subset males is slightly higher than subset females. As expected, the average years of employment for subset 1 and subset 2 are 6.19 and 9.01 which are both higher than the average in dataset 5.12 as we filtered based on this criteria. The average for subset males is higher than subset females. The average hourly wage for subset 1 and subset 2 are 5.34 and 8.94. Thhe average of total dataset is 5.99 which lower than the male subset average but higher than the female subset average. This hints towards a wage differential based on gender which we will try to investigate with difference of means test in the next question.

Difference in Means Test

Conduct a statistical test of the significance of the difference between the means of two meaningful subsets of the data and interpret the results.

In [195…

```
# If variances are similar, we can use an independent t-test, but if not, we have to use a Welch's t-test
# So we first need to see if variances are similar.

from scipy.stats import levene

# H0: variances in males and females with high education and high experience levels are equal
# H1: variances in males and females with high education and high experience levels are not equal
```

```
# Doing the levene test for variances
levene_value, p_value_levene = levene(subset_male_highedu_highexp['hourly_wage'], subset_female_highedu_highexp['hourly_wage'])

alpha = 0.05
# assuming significance level of 0.05 (defined as alpha)
if p_value_levene<alpha:
    print(f'P-Value is {p_value_levene} (less than alpha) hence reject null')
else:
    print(f'P-Value is {p_value_levene} (greater than alpha) hence do not reject null')
```

```
P-Value is 5.370382393448693e-05 (less than alpha) hence reject null
```

In [196…

```
#since we rejected the null hypothesis in earlier levene test, we need to use Welch t-test of unequal variances

#H0: Female get same wages as Males at high education and high experience levels
#H1: Female get lower wages as Males at high education and high experience levels


# Extract wage values
female = subset_female_highedu_highexp['hourly_wage']
male   = subset_male_highedu_highexp['hourly_wage']

# T-test (Welch)
t_stat, p_val = ttest_ind(female, male, equal_var=False)

# Compute means, stds, and sizes
mean_f = female.mean()
mean_m = male.mean()
std_f = female.std()
std_m = male.std()
n_f = female.count()
n_m = male.count()


# Print results
print("------ Difference in Means Test (Female vs Male) ------")
print(f"Female (high education + high experience): mean = {mean_f:.3f}, std = {std_f:.3f}, n = {n_f}")
print(f"Male (high education + high experience):   mean = {mean_m:.3f}, std = {std_m:.3f}, n = {n_m}")
print("\nT-statistic:", round(t_stat, 4))
print("P-value:", p_val)
```

```
------ Difference in Means Test (Female vs Male) ------
Female (high education + high experience): mean = 5.343, std = 2.828, n = 117
Male (high education + high experience):   mean = 8.940, std = 4.503, n = 139

T-statistic: -7.7722
P-value: 2.367900820163561e-13
```

Interpretation: A Welch's two-sample t-test was used to assess the difference in mean hourly wages between females and married who have both high education and high experience. Females earned an average of £5.34 (SD = 2.83, n = 117), compared with £8.94 for males (SD = 4.50, n = 139). The test produced a t-statistic of –7.77 and a p-value of 2.37e-13. Since the p-value is far below the 0.05 significance level, the null hypothesis of equal means is rejected. This provides very strong evidence that males earn significantly higher wages than females at similar high education level and high experience level.

Grouping and Summary Statistics

Perform a meaningful data grouping by a certain categorical variable and display summarized information for each group (e.g., the mean or sum within the group).

In [197…

```python
# Group by race and compute summary statistics
group_race = df.groupby('race').agg({
    'hourly_wage': ['mean', 'median', 'std', 'min', 'max'],
    'years_in_education': ['mean'],
    'years_in_employment': ['mean'],
    'num_dependents': ['mean', 'median'],
    'married': ['mean']
})

group_race
```

Out[197…

	hourly_wage					years_in_education	years_in_employment	num_dependents		married
	mean	median	std	min	max	mean	mean	mean	median	mean
race										
nonwhite	5.576296	4.585	3.095026	1.96	15.00	11.870370	5.388889	1.333333	1.0	0.518519
white	5.935690	4.670	3.735612	0.53	24.98	12.632696	5.084926	1.010616	1.0	0.622081

Intrepretation: A meaningful grouping was performed using race as the categorical variable. The results show that white individuals earn slightly higher hourly wages on average (£5.94 vs £5.58) and also exhibit a wider wage range, with much higher maximum earnings than nonwhite individuals. White individuals also have higher educational attainment, averaging 12.63 years compared with 11.87 years for nonwhite individuals. Although nonwhite individuals have slightly more years of employment experience, this does not translate into higher wages, indicating that additional factors may influence earnings. The nonwhite group also tends to have more dependents and a lower proportion of married individuals. Overall, grouping by race highlights notable socioeconomic differences across demographic categories within the dataset.

Linear Regression Analysis

Conduct a linear regression analysis and interpret its model output including its accuracy

In [198…

```
# linear regression predicting hourly wage
model_full = sm.OLS.from_formula(
    formula="hourly_wage ~ years_in_education + years_in_employment + num_dependents + married + C(gender) + C(race)",
    data=df
).fit()

print(model_full.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            hourly_wage   R-squared:                       0.348
Model:                            OLS   Adj. R-squared:                  0.341
Method:                 Least Squares   F-statistic:                     46.17
Date:                Thu, 11 Dec 2025   Prob (F-statistic):           2.71e-45
Time:                        19:37:26   Log-Likelihood:                -1315.1
No. Observations:                 525   AIC:                             2644.
Df Residuals:                     518   BIC:                             2674.
Df Model:                           6
Covariance Type:            nonrobust
=======================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
Intercept              -2.8196      0.761     -3.705      0.000      -4.315      -1.324
C(gender)[T.male]       1.6918      0.269      6.293      0.000       1.164       2.220
C(race)[T.white]       -0.0140      0.432     -0.032      0.974      -0.863       0.835
years_in_education      0.5191      0.049     10.531      0.000       0.422       0.616
years_in_employment     0.1506      0.019      7.945      0.000       0.113       0.188
num_dependents          0.0983      0.109      0.904      0.366      -0.115       0.312
married                 0.7393      0.283      2.610      0.009       0.183       1.296
==============================================================================
Omnibus:                      208.236   Durbin-Watson:                   1.814
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              903.685
Skew:                           1.758   Prob(JB):                    5.85e-197
Kurtosis:                       8.380   Cond. No.                         86.8
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Interpretation: The full multivariate regression model shows that hourly wage is significantly influenced by levels of education, employment experience, gender, and marital status. Both education and employment experience have positive and statistically significant effects, consistent with economic theory. Males earn significantly higher hourly wages than females even after controlling for multiple demographic and human-capital variables, indicating the presence of a gender wage gap in the dataset. Marital status also shows a positive association with wages, while race and number of dependents do not have statistically significant effects when other variables are included. The model explains approximately 34% of wage variation, demonstrating moderate predictive accuracy given the limited set of variables. Overall, the regression provides meaningful insight into how human capital and demographic factors jointly influence wage outcomes.

In [199…

```
# Lets check the multicollinearity between the independant variables
formula = "hourly_wage ~ years_in_education + years_in_employment + num_dependents + married + C(gender) + C(race)"

# Build design matrices
y, X = dmatrices(formula, data=df, return_type='dataframe')

# Ensure everything is numeric (float)
X = X.astype(float)

# Compute VIF
vif_data = pd.DataFrame()
vif_data["Variable"] = X.columns
vif_data["VIF Value"] = [
    variance_inflation_factor(X.values, i) for i in range(X.shape[1])
]

print(vif_data)
```

```
              Variable  VIF Value
0            Intercept  34.179092
1    C(gender)[T.male]   1.063427
2     C(race)[T.white]   1.017401
3   years_in_education   1.081753
4  years_in_employment   1.104218
5       num_dependents   1.091978
6              married   1.125191
```

Variables with a VIF between 1-5 are generally acceptable and hence are retained in the model. We can safely say that that there is no high multi-collinearity hence we will not drop variables based on this. However, we will drop the variables number of dependants and race since they have a p value > 0.05.

In [200…

```
# The model below represents the final linear regression model
# Variables with p-value>0.05 have been excluded, as mentioned above
# The hourly wage variable has also been dropped - that's because it is an outcome.

final_model = sm.OLS.from_formula(
    formula="hourly_wage ~ years_in_education + years_in_employment + married + C(gender)",
    data=df).fit()

print(final_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            hourly_wage   R-squared:                       0.347
Model:                            OLS   Adj. R-squared:                  0.342
Method:                 Least Squares   F-statistic:                     69.21
Date:                Thu, 11 Dec 2025   Prob (F-statistic):           5.85e-47
Time:                        19:37:26   Log-Likelihood:                -1315.6
No. Observations:                 525   AIC:                             2641.
Df Residuals:                     520   BIC:                             2662.
Df Model:                           4
Covariance Type:            nonrobust
=======================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
Intercept              -2.6165      0.632     -4.140      0.000      -3.858      -1.375
C(gender)[T.male]       1.6889      0.268      6.292      0.000       1.162       2.216
years_in_education      0.5085      0.048     10.643      0.000       0.415       0.602
years_in_employment     0.1493      0.019      7.912      0.000       0.112       0.186
married                 0.7845      0.278      2.827      0.005       0.239       1.330
==============================================================================
Omnibus:                      208.392   Durbin-Watson:                   1.819
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              906.638
Skew:                           1.758   Prob(JB):                    1.34e-197
Kurtosis:                       8.392   Cond. No.                         69.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Interpretation: A comparison of the full regression model and a reduced model shows that removing the statistically insignificant predictors (race and number of dependents) does not reduce the explanatory power of the model. Both models have nearly identical R^2 values (~0.35), but the reduced model achieves a slightly higher adjusted R^2, lower AIC and BIC, and a lower condition number, indicating a more stable and parsimonious specification. All remaining predictors—gender, education, employment experience, and marital status—remain significant. Therefore, the reduced model is preferred, as it provides an equally strong fit with fewer predictors and improved interpretability.

In [201…

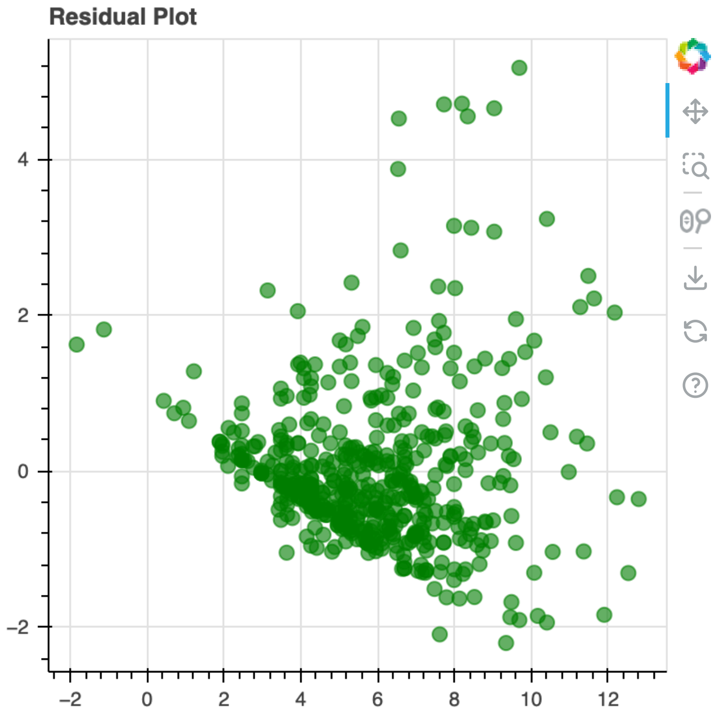
```
# testing linearity assumptions
from bokeh.io import output_notebook
output_notebook()
from bokeh.plotting import figure
from bokeh.io.export import export_png
from bokeh.io import show
```

BokehJS 3.6.2 successfully loaded.

In [202…

```
# the x axis is the fitted values
# the y axis is the standardized residuals
```

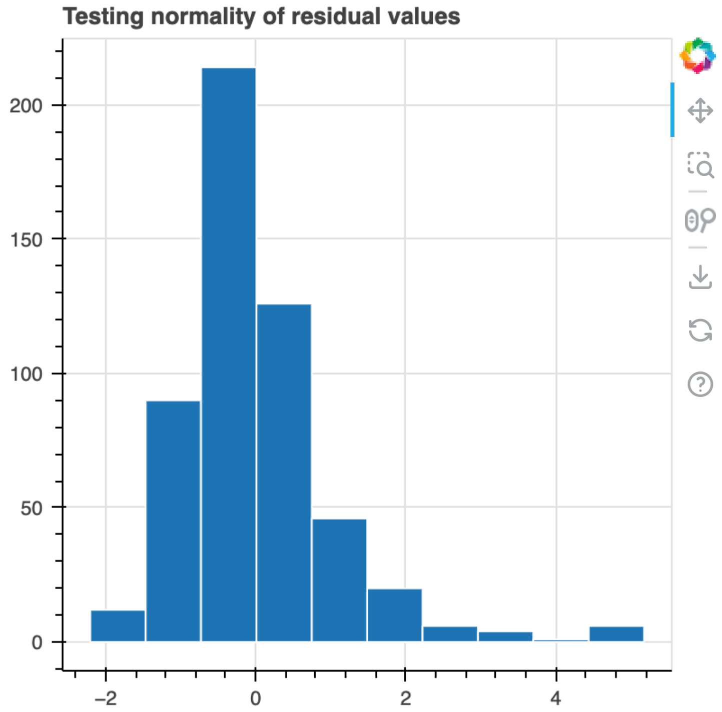
```
st_resids = final_model.get_influence().resid_studentized_internal
fig = figure(height=400, width=400, title="Residual Plot")
fig.circle(final_model.fittedvalues, st_resids, size=8, color="green", alpha=0.6)
show(fig)
```



Interpretation: The residual plot shows the standardized residuals plotted against the fitted values from the regression model. The points appear scattered without a strong visible pattern, suggesting that the assumption of linearity is reasonably satisfied. However, there is some mild funneling, with residuals spreading out slightly as fitted values increase, indicating possible heteroscedasticity (non-constant variance). A few residuals lie above 3 or below –2, suggesting the presence of some outliers, but not enough to invalidate the model. Overall, the residual plot indicates that the model fits reasonably well, though the slight increase in variance at higher fitted values suggests that the assumption of constant variance may not hold perfectly.

In [203…

```
#testing normality of error values
hist, edges = np.histogram(st_resids, bins=10)
fig = figure(height=400, width=400, title = 'Testing normality of residual values')
fig.quad(top=hist, bottom=0, left=edges[:-1], right=edges[1:], line_color="white")
show(fig)
```



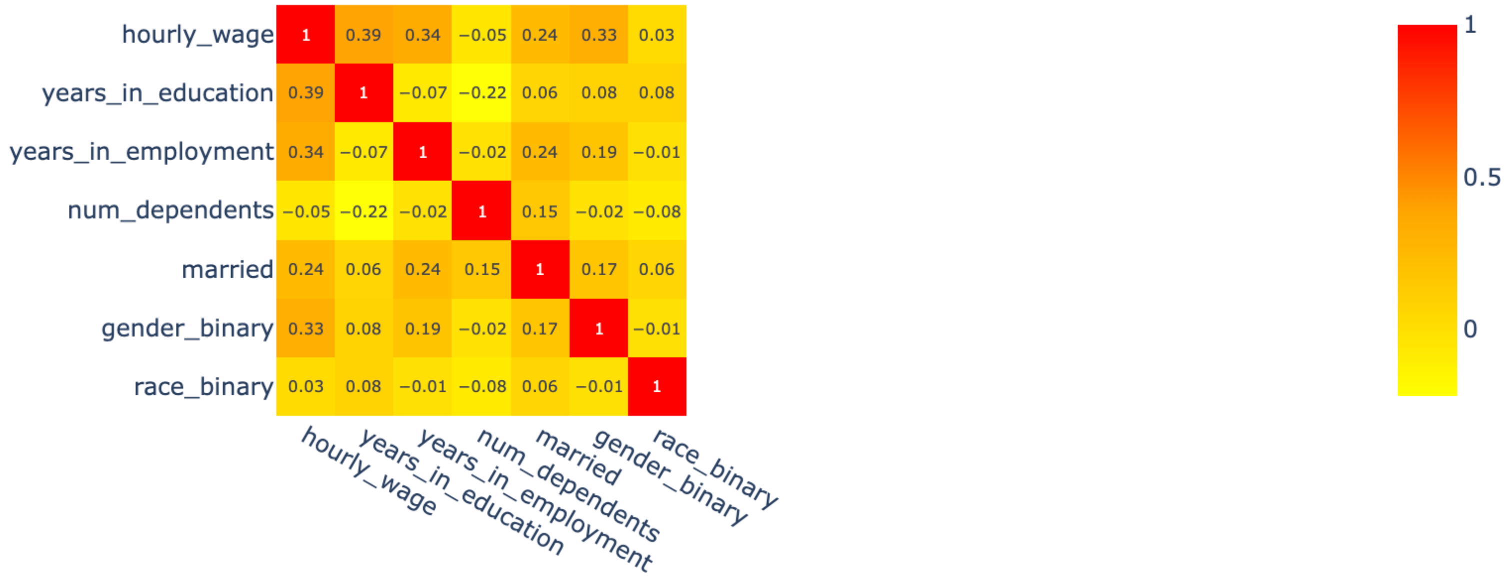
Residuals are approximately but not perfectly normal, which is common in wage data. The histogram of residuals and the positive skew suggest that a small number of high-wage observations introduce non-normality. The Jarque–Bera test confirms this, with a p-value well below 0.05, indicating that the residuals deviate significantly from a normal distribution. However, mild violations of normality and heteroscedasticity typically do not greatly affect the accuracy of predicted values in large samples. Therefore, despite these assumption breaches, the model remains reasonable and useful for interpreting the main relationships between predictors and hourly wage.FINAL MODEL Hourly_wage = –2.62 + 1.69*gender + 0.51*years_in_education + 0.15*years_in_employment + 0.78*married Gender = 1 if the individual is male, 0 if female Years in Education = total years of education completed Years in Employment = cumulative years of work experience Married = 1 if married, 0 if not married

In [204…

```
# plotly is used to visually see the strength of correaltion between all the selected variables for better analysis
# This allows us to see which variables are stringly related to each other (and which are strongly related with hourly_wage)

#convert string variables to binary
df['gender_binary'] = df['gender'].map({'female': 0, 'male': 1})
df['race_binary'] = df['race'].map({'nonwhite': 0, 'white': 1})

corr_matrix = round(df[
['hourly_wage', 'years_in_education', 'years_in_employment', 'num_dependents', 'married' , 'gender_binary' , 'race_binary']].corr(
custom_color_scale = [[0,
'yellow'], [0.5,
'orange'], [1,
'red']]
fig = px.imshow(corr_matrix, text_auto=True, color_continuous_scale=custom_color_scale)
fig.show()
```



Interpretation: The heatmap shows that education, employment experience, and gender are the variables more strongly related to hourly wage with moderately high positive correlations: 0.39, 0.34 and 0.33 respectively. Marital status has also a moderaly positive correlation with hourly wage at 0.22. Race and number of dependents show almost no correlation. These findings are fully consistent with the regression analysis, strengthening confidence in the model's conclusions. Other insights include: There is a moderate positive correlation (0.24) between employment experience and marital status, likely due to older individuals having both longer work histories and higher marriage rates. It is meaningful enough to suggest demographic overlap between employment experience and marital patterns. The moderate negative correlation between education and number of dependents (–0.22) indicates that individuals with more education tend to have fewer dependents. These relationships may reflect life-cycle or socioeconomic factors influencing family size and educational attainment.

Conclusion

This exploratory analysis provided a structured examination of hourly wage patterns using descriptive statistics, visualisations, hypothesis testing and regression modelling. The investigation highlighted clear differences in wages across demographic and human-capital factors. Gender showed statistically significant association with wage levels, and the difference-in-means test confirmed a substantial wage gap between men and women at similar education and experience levels. The full regression model demonstrated that education and employment experience are strong positive predictors of hourly wage, while gender and marital status also remains significant even after controlling for other variables. Race and number of dependents were not significant predictors once all factors were considered. The model explained around one-third of wage variation, reflecting moderate predictive accuracy and the influence of additional factors not captured in the dataset. Overall, the analysis offered meaningful insight into how demographic and human-capital characteristics shape wage outcomes, while reinforcing key data-analysis skills such as data cleaning, visualisation, statistical testing and model interpretation.

Declaration:

This is to certify that the work I am submitting is my own, and I am aware of Aston University's regulations concerning plagiarism and collusion. No substantial part of this work has been submitted by me for any other accredited assessment, and I acknowledge that doing so may lead to a reduction in marks. AI tools (specifically Gemini and ChatGPT) were used only to support my work by editing, proofreading, refining explanations, and providing example code patterns or stylistic guidance. All analytical decisions, coding logic, data exploration, and interpretations were independently produced and adapted by me for the specific requirements of this report.